

Name \_\_\_\_\_ Date \_\_\_\_\_ Period \_\_\_\_\_

## Domain and Range: All Three Notations

Algebra II Honors | Unit 1 | Day 1 | TEK A2.7.I

**Objective:** I can write domain and range in verbal, inequality, set, and interval notation, determine a domain algebraically from a function rule, and use union notation for a domain that comes in separate pieces.

### Vocabulary

**Set-builder notation** —  $\{x \mid \text{condition}\}$ , read "the set of all  $x$  \_\_\_\_\_ the condition holds."

**Interval notation** — \_\_\_\_\_ includes an endpoint, \_\_\_\_\_ excludes it. Infinity always takes a \_\_\_\_\_.

**Union  $\cup$**  — joins two separate intervals when the domain has a \_\_\_\_\_ in it.

**Undefined** — an input the rule cannot accept. Two causes in this course: a \_\_\_\_\_ and a \_\_\_\_\_.

### Part 1 — From Pairs and Tables

List least to greatest, each value once.

a.  $\{(-4, -6), (0, -1), (2, 4), (3, -7), (7, 7)\}$  D \_\_\_\_\_ R \_\_\_\_\_

b.  $\{(-5, 3), (-1, 0), (2, 3), (6, -8)\}$  D \_\_\_\_\_ R \_\_\_\_\_

c. Why does the range in (b) have three values when there are four pairs? \_\_\_\_\_

### Part 2 — All Three Notations

Complete every cell.

| Verbal description                      | Inequality | Set notation | Interval notation |
|-----------------------------------------|------------|--------------|-------------------|
| all real numbers                        | _____      | _____        | _____             |
| all reals at least -6                   | _____      | _____        | _____             |
| all reals less than 5                   | _____      | _____        | _____             |
| all reals from -3 to 3, 3 included only | _____      | _____        | _____             |
| all reals except 2                      | _____      | _____        | _____             |
| just the numbers -2, 0, and 4           | (none)     | _____        | _____             |

The last two rows are the ones to think hardest about. Explain the difference between them:

\_\_\_\_\_

Row 5 is continuous with one point removed; row 6 is \_\_\_\_\_

### Part 3 — Finding a Domain from the Rule

No graph, no table - reason from the algebra. Ask what the rule forbids, then exclude it.

| Function                | What is forbidden? | Inequality | Interval notation |
|-------------------------|--------------------|------------|-------------------|
| $f(x) = \sqrt{x - 3}$   | _____              | _____      | _____             |
| $g(x) = 1/(x - 5)$      | _____              | _____      | _____             |
| $h(x) = \sqrt{2x + 8}$  | _____              | _____      | _____             |
| $k(x) = 1/\sqrt{x - 1}$ | _____              | _____      | _____             |

Look at  $k(x)$ . Why is the answer  $x > 1$  and not  $x \geq 1$ ?

At  $x = 1$  the radicand is 0, and that would make the denominator \_\_\_\_\_ — so 1 must be excluded too.

#### Part 4 — A Domain in Three Pieces

Find the domain of  $m(x) = (x + 2)/(x^2 - 9)$ . Show the factoring.

Denominator factors as \_\_\_\_\_, so  $x$  cannot be \_\_\_\_\_

Domain, interval notation: \_\_\_\_\_

Domain, set notation: \_\_\_\_\_

Which notation communicates this domain more efficiently, and why?

#### Part 5 — Going Further

1. Explain why  $\infty$  can never take a bracket, no matter the function.

2. Write a function whose domain is exactly  $[4, \infty)$ . \_\_\_\_\_

3. Write a function whose domain is exactly  $(-\infty, 0) \cup (0, \infty)$ . \_\_\_\_\_

4. Can a domain be empty? Give a rule with no valid inputs, or argue that none exists.

Sample: \_\_\_\_\_

#### Exit Ticket

Give the domain of  $f(x) = \sqrt{x + 5}/(x - 2)$  in interval notation, and justify each restriction.

Domain \_\_\_\_\_

Two reasons: \_\_\_\_\_

## Domain and Range from Problem Situations

Algebra II Honors | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

**Objective:** I can build a function from a described situation, determine the domain the situation allows versus the domain the equation allows, and interpret both in context.

### Vocabulary

**Function domain** — every input the \_\_\_\_\_ accepts.    **Contextual domain** — every input the \_\_\_\_\_ allows.

**Constraint** — a condition the situation imposes that the algebra does not, such as "a length must be \_\_\_\_\_."

**Average rate of change** from  $a$  to  $b$  = \_\_\_\_\_ — the slope between two points.

### Part 1 — Build the Function First

*You are given the situation, not the equation. Write the rule, then the domain.*

- a. A rectangle has perimeter 40 ft. One side is  $x$ . Write its area as a function of  $x$ .

\_\_\_\_\_

\_\_\_\_\_

Other side = \_\_\_\_\_     $A(x)$  = \_\_\_\_\_    Contextual domain = \_\_\_\_\_

- b. Why is the domain open at both ends rather than closed? \_\_\_\_\_

- c. What domain does the equation  $x(20 - x)$  accept on its own? \_\_\_\_\_

### Part 2 — Three Situations

*Write the rule, the contextual domain, and the range. Use interval or set notation as fits.*

1. A taxi charges \$4 plus \$2.50 per mile, for trips up to 12 miles.

$C(m)$  = \_\_\_\_\_    Domain \_\_\_\_\_    Range \_\_\_\_\_

2. A club sells shirts for \$15. Each shirt costs them \$6, and the screen fee is \$90 total.

\_\_\_\_\_

\_\_\_\_\_

Profit  $P(n)$  = \_\_\_\_\_    Break-even at  $n$  = \_\_\_\_\_

Domain \_\_\_\_\_    Discrete or continuous? \_\_\_\_\_

3. A ball is thrown upward:  $h(t) = -16t^2 + 48t + 4$  feet, until it lands.

\_\_\_\_\_

\_\_\_\_\_

Domain: from  $t = 0$  until  $h = 0$ , about \_\_\_\_\_    Why can  $t$  not be negative?

\_\_\_\_\_

The range starts at  $h$  = \_\_\_\_\_ and the maximum is at  $t = 1.5$ , giving  $h$  = \_\_\_\_\_, so range

\_\_\_\_\_

### Part 3 — Average Rate of Change

Use the taxi function  $C(m) = 4 + 2.5m$  and the ball function  $h(t)$  from above.

- a. Average rate of change of  $C$  from  $m = 2$  to  $m = 8$ :

\_\_\_\_\_

= \_\_\_\_\_ Why is this the same for every interval? \_\_\_\_\_

- b. Average rate of change of  $h$  from  $t = 0$  to  $t = 1$ , then from  $t = 2$  to  $t = 3$ :

\_\_\_\_\_

\_\_\_\_\_

First: \_\_\_\_\_ Second: \_\_\_\_\_ What does the sign change mean?

\_\_\_\_\_

### Part 4 — Going Further

1. Describe a real situation whose contextual domain is exactly the whole numbers 0 to 30.

\_\_\_\_\_

\_\_\_\_\_

2. Give a situation where the range is bounded but the domain is not.

\_\_\_\_\_

Sample: \_\_\_\_\_

3. Can a contextual domain ever be LARGER than the function domain? Explain.

\_\_\_\_\_

No — \_\_\_\_\_

### Exit Ticket

A square sheet of metal 24 cm on a side has a square of side  $x$  cut from each corner. Write the volume of the resulting open box and give the contextual domain, with justification.

\_\_\_\_\_

\_\_\_\_\_

$V(x) =$  \_\_\_\_\_ Domain \_\_\_\_\_

Justify the upper bound: \_\_\_\_\_

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## Continuous vs. Discrete

Algebra II Honors | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

**Objective:** I can classify a domain as continuous or discrete, write each with the notation that fits, recognize step behavior, and judge when discreteness is a property of the quantity versus a modeling choice.

### Vocabulary

**Continuous** — every value between any two points belongs to the domain. **Discrete** — the domain is \_\_\_\_\_.

**Step function** — a \_\_\_\_\_ domain paired with a \_\_\_\_\_ range. The output jumps.

**Sequence** — a function whose domain is the \_\_\_\_\_. Always discrete.

### Part 1 — Classify and Justify

Write C or D, then give the domain in correct notation and one sentence of justification.

| Situation                         | C/D   | Domain | Why   |
|-----------------------------------|-------|--------|-------|
| Distance a car travels in 3 hours | _____ | _____  | _____ |
| Number of students on 6 buses     | _____ | _____  | _____ |
| Mass of water in a beaker         | _____ | _____  | _____ |
| Term number in a sequence         | _____ | _____  | _____ |

### Part 2 — One Equation, Two Domains

A cube-shaped box of side  $x$  has volume  $V(x) = x^3$ .

|             |   |   |    |    |
|-------------|---|---|----|----|
| <b>x</b>    | 1 | 2 | 3  | 4  |
| <b>V(x)</b> | 1 | 8 | 27 | 64 |

Any size box: domain \_\_\_\_\_ range \_\_\_\_\_

Built from 1-inch blocks: domain \_\_\_\_\_ range \_\_\_\_\_

The algebra did not change. What did? \_\_\_\_\_

### Part 3 — Step Behavior

A parking garage charges \$3 for each hour or any part of an hour, up to 4 hours. Park for 1.2 hours and you pay for 2.

|                  |     |     |     |     |     |     |
|------------------|-----|-----|-----|-----|-----|-----|
| <b>t (hours)</b> | 0.5 | 1.0 | 1.2 | 2.0 | 2.7 | 3.5 |
| <b>Cost</b>      |     |     |     |     |     |     |

Domain (time parked): \_\_\_\_\_ Range (cost): \_\_\_\_\_

This is the case the on-level notes do not cover: a domain and range that disagree. Explain how that happens.

\_\_\_\_\_  
 \_\_\_\_\_

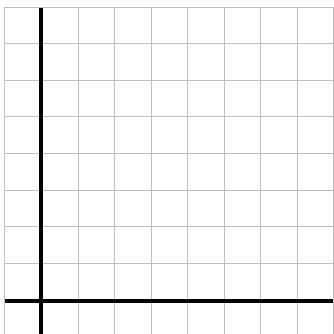
Time varies smoothly, but the \_\_\_\_\_ at each whole hour instead of rising gradually.

Why must  $t = 1.0$  cost \$3 and not \$6? \_\_\_\_\_

#### Part 4 — Graph Both Kinds

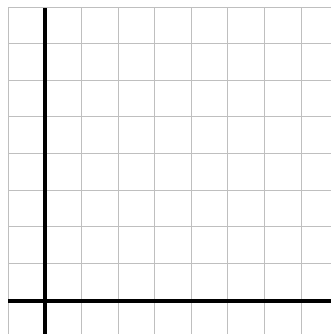
Each square is 1 unit. Use open and closed dots carefully.

- A. Discrete.** A tutor charges \$2 per session, up to 4 sessions. **B. Continuous.** A candle burns 2 cm per hour from 8 cm.



Domain \_\_\_\_\_

Range \_\_\_\_\_



Rule  $h =$  \_\_\_\_\_

Domain \_\_\_\_\_ Range \_\_\_\_\_

#### Part 5 — Going Further

1. Is money continuous or discrete? Argue both sides, then commit to one.

\_\_\_\_\_  
\_\_\_\_\_

Discrete in practice \_\_\_\_\_, often modeled as continuous because the steps are tiny.

2. Give a situation with a discrete domain and a continuous range, or argue it is impossible.

\_\_\_\_\_  
\_\_\_\_\_

Impossible — \_\_\_\_\_

#### Exit Ticket

Postage costs \$0.60 for the first ounce and \$0.24 for each additional ounce or part of an ounce, up to 5 ounces.

Domain (weight) \_\_\_\_\_ Range (cost) \_\_\_\_\_

Classify the domain and the range, and explain why they differ: \_\_\_\_\_

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## Parent Functions I: $\sqrt{x}$ and $|x|$

Algebra II Honors | Unit 1 | Day 4 | TEK A2.2.A

**Objective:** I can graph and analyze the square root and absolute value parent functions, write absolute value as a piecewise rule, and explain why the two behave differently for negative inputs.

### Vocabulary

**Parent function** — the simplest member of a family, with nothing added or multiplied.

**Piecewise rule** — one function defined by \_\_\_\_\_ on different parts of the domain.

**Principal square root** —  $\sqrt{x}$  means the \_\_\_\_\_ root only, which is why  $\sqrt{x}$  is a function.

### Part 1 — Build Both Tables

Fill both. Note where outputs are undefined and where they repeat.

| x                 | -4 | -1 | 0 | 1 | 4 | 9 |
|-------------------|----|----|---|---|---|---|
| $f(x) = \sqrt{x}$ |    |    |   |   |   |   |

| x            | -4 | -1 | 0 | 1 | 4 | 9 |
|--------------|----|----|---|---|---|---|
| $g(x) =  x $ |    |    |   |   |   |   |

One table has undefined entries, the other has repeats. Explain what each tells you:

\_\_\_\_\_

\_\_\_\_\_

$\sqrt{x}$  has a restricted \_\_\_\_\_;  $|x|$  is not \_\_\_\_\_, which will matter on Day 12.

### Part 2 — Absolute Value as a Piecewise Rule

Absolute value is distance from zero, so it never returns a negative. Written as two cases:

$|x| =$  \_\_\_\_\_ when  $x \geq 0$ , and  $|x| =$  \_\_\_\_\_ when  $x < 0$ .

The second case looks like it produces a negative. Explain why it does not:

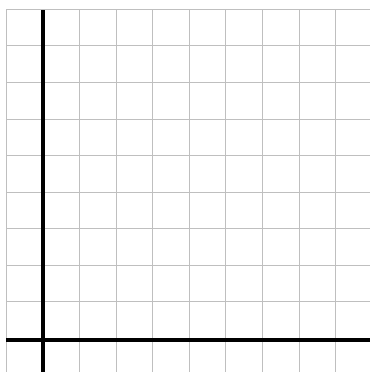
\_\_\_\_\_

If  $x$  is already negative, \_\_\_\_\_. For example  $-(-5) = 5$ .

### Part 3 — Graph Them Both

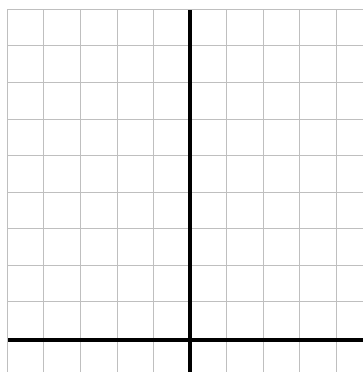
Each square is 1 unit. Plot from your tables, then connect smoothly.

A.  $f(x) = \sqrt{x}$   $x$  from 0 to 9



Domain \_\_\_\_\_ Range \_\_\_\_\_

B.  $g(x) = |x|$   $x$  from -5 to 5



Domain \_\_\_\_\_ Range \_\_\_\_\_

## Part 4 — Solution Counts

Solve each. The number of answers is the point, not just the answers.

1.  $|x| = 6 \rightarrow x =$  \_\_\_\_\_ how many? \_\_\_\_\_ 2.  $\sqrt{x} = 6 \rightarrow x =$  \_\_\_\_\_ how many? \_\_\_\_\_

3.  $|x| = 0 \rightarrow x =$  \_\_\_\_\_ how many? \_\_\_\_\_ 4.  $|x| = -3 \rightarrow$  \_\_\_\_\_

Item 4 has no solution. What does that say about the range of  $|x|$ ? \_\_\_\_\_

Connect item 1 to the graph: why do two  $x$ -values give the same output?

\_\_\_\_\_

The V is symmetric about the \_\_\_\_\_, so a horizontal line meets it twice.

## Part 5 — Going Further

1. Give the domain of each, in interval notation.

$\sqrt{x-7}$  \_\_\_\_\_  $\sqrt{3x+6}$  \_\_\_\_\_  $|x-7|$  \_\_\_\_\_

2. Why does the absolute value one have no restriction while the radicals do?

\_\_\_\_\_

3. On the interval  $[1, 9]$ , give the minimum and maximum of  $\sqrt{x}$ . min \_\_\_\_\_ max \_\_\_\_\_

4. Is  $\sqrt{x^2}$  the same function as  $|x|$ ? Test  $x = -5$  and  $x = 5$ , then decide.

\_\_\_\_\_

\_\_\_\_\_

Yes — \_\_\_\_\_

## Exit Ticket

Both parent functions have minimum 0 and pass through the origin.

Name one attribute that separates them: \_\_\_\_\_ and one they share besides those two: \_\_\_\_\_

Write  $|x|$  as a piecewise rule from memory: \_\_\_\_\_



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## Parent Functions II: $x^3$ and $\sqrt[3]{x}$

Algebra II Honors | Unit 1 | Day 5 | TEK A2.2.A

**Objective:** I can graph and analyze the cubic and cube root parent functions, prove algebraically that each is an odd function, and describe how their rates of change differ.

### Vocabulary

**Odd function** — satisfies \_\_\_\_\_ for every  $x$ . Its graph has \_\_\_\_\_ symmetry about the origin.

**Even function** — satisfies \_\_\_\_\_. Its graph is symmetric across the \_\_\_\_\_.

**Average rate of change** on  $[a, b]$  = \_\_\_\_\_

### Part 1 — Two Tables, One Relationship

| $x$          | -8 | -2 | -1 | 0 | 1 | 2 | 8 |
|--------------|----|----|----|---|---|---|---|
| $f(x) = x^3$ |    |    |    |   |   |   |   |

| $x$                  | -8 | -2 | -1 | 0 | 1 | 2 | 8 |
|----------------------|----|----|----|---|---|---|---|
| $g(x) = \sqrt[3]{x}$ |    |    |    |   |   |   |   |

Find a pair of entries showing the two functions undo each other: \_\_\_\_\_

### Part 2 — Prove Both Are Odd

*Substitute  $-x$  into each rule and simplify. Do not just check one number.*

a.  $f(x) = x^3$ . Find  $f(-x)$ :

\_\_\_\_\_  
\_\_\_\_\_

$f(-x) =$  \_\_\_\_\_ so  $f$  is \_\_\_\_\_

b.  $g(x) = \sqrt[3]{x}$ . Find  $g(-x)$ :

\_\_\_\_\_  
\_\_\_\_\_

$g(-x) =$  \_\_\_\_\_ so  $g$  is \_\_\_\_\_

c. Test  $h(x) = x^2$  the same way. Is it odd, even, or neither? \_\_\_\_\_

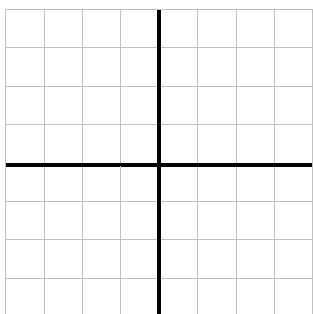
d. What does "odd" tell you about the graph without graphing it?

\_\_\_\_\_

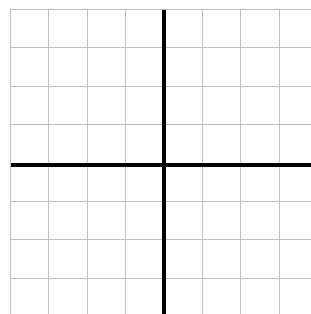
### Part 3 — Graph Them Both

*Both grids run -8 to 8 on each axis, so 1 square = 2 units.*

A.  $f(x) = x^3$



B.  $g(x) = \sqrt[3]{x}$



Domain \_\_\_\_\_ Range \_\_\_\_\_

Mirror line between them: \_\_\_\_\_

## Part 4 — Rates of Change

Compute the average rate of change of  $f(x) = x^3$  on each interval.

On  $[0, 1]$ : \_\_\_\_\_ On  $[1, 2]$ : \_\_\_\_\_ On  $[2, 3]$ : \_\_\_\_\_

Now  $g(x) = \sqrt[3]{x}$  on  $[0, 1]$ : \_\_\_\_\_ On  $[1, 8]$ : \_\_\_\_\_

Describe the contrast in one sentence:

\_\_\_\_\_

The cubic \_\_\_\_\_ as  $x$  grows while the cube root \_\_\_\_\_ — which is what makes them mirror images.

## Part 5 — Going Further

1. Solve  $x^3 = 125$  and  $x^2 = 25$ . Count the real solutions for each.

$x^3 = 125 \rightarrow$  \_\_\_\_\_  $x^2 = 25 \rightarrow$  \_\_\_\_\_

2. Explain the difference using odd and even.

\_\_\_\_\_

\_\_\_\_\_

$x^3$  is odd and never repeats an output;  $x^2$  is even, so \_\_\_\_\_

3. Which of the four parent functions so far accept negative inputs? \_\_\_\_\_

4. Can a function be both even and odd? Find one or prove none exists.

\_\_\_\_\_

Yes — \_\_\_\_\_

## Exit Ticket

A function has domain and range all reals, passes through the origin, and satisfies  $f(-x) = -f(x)$ .

Name two parent functions this could be: \_\_\_\_\_

Prove your choice is odd, algebraically: \_\_\_\_\_

Name \_\_\_\_\_ Date \_\_\_\_\_ Period \_\_\_\_\_

## Parent Functions III: $1/x$ and Asymptotes

Algebra II Honors | Unit 1 | Day 6 | TEK A2.2.A

**Objective:** I can graph the reciprocal parent function, describe its end behavior precisely, prove it is odd, and locate vertical asymptotes of related rules from the denominator.

### Vocabulary

**Vertical asymptote** — a vertical line the graph \_\_\_\_\_, located where the function is \_\_\_\_\_.

**Horizontal asymptote** — describes \_\_\_\_\_. Unlike a vertical one, it \_\_\_\_\_ be crossed by other functions.

**End behavior notation** — "as  $x \rightarrow \infty$ ,  $f(x) \rightarrow 0$ " is read: as  $x$  grows \_\_\_\_\_, the output approaches 0.

### Part 1 — Build the Table

| $x$          | -4 | -2 | -1 | -1/2 | 0 | 1/2 | 1 | 2 | 4 |
|--------------|----|----|----|------|---|-----|---|---|---|
| $f(x) = 1/x$ |    |    |    |      |   |     |   |   |   |

Why is  $x = 0$  excluded, and what is the domain in set notation?

Division by zero is undefined, so \_\_\_\_\_

### Part 2 — End Behavior, Precisely

Complete both patterns, then write the behavior in words.

| $x$   | 1 | 0.1 | 0.01 | 0.001 |
|-------|---|-----|------|-------|
| $1/x$ |   |     |      |       |

| $x$   | 1 | 10 | 100 | 1000 |
|-------|---|----|-----|------|
| $1/x$ |   |    |     |      |

As  $x \rightarrow 0$  from the right,  $f(x) \rightarrow$  \_\_\_\_\_ As  $x \rightarrow 0$  from the left,  $f(x) \rightarrow$  \_\_\_\_\_

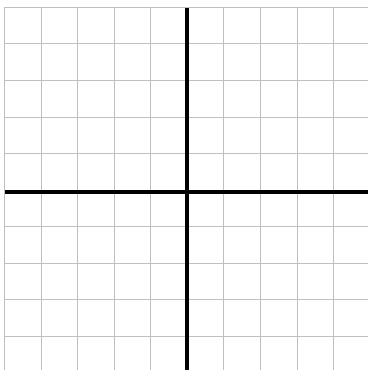
As  $x \rightarrow \infty$ ,  $f(x) \rightarrow$  \_\_\_\_\_ As  $x \rightarrow -\infty$ ,  $f(x) \rightarrow$  \_\_\_\_\_

So the asymptotes are  $x =$  \_\_\_\_\_ and  $y =$  \_\_\_\_\_. Does the graph ever reach  $y = 0$ ? \_\_\_\_\_

### Part 3 — Graph and Prove

Each square is 1 unit. Two branches - do not connect them across the  $y$ -axis.

$f(x) = 1/x$ , with both asymptotes dashed.



Prove  $f$  is odd.

$f(-x) =$  \_\_\_\_\_

so  $f(-x) =$  \_\_\_\_\_

Quadrants used: \_\_\_\_\_

Intercepts: \_\_\_\_\_

Max or min: \_\_\_\_\_

Why does having no intercepts follow from the asymptotes?

\_\_\_\_\_

The axes \_\_\_\_\_ the asymptotes, and a graph never reaches its asymptotes.

## Part 4 — Asymptotes of Related Rules

*Reason from the denominator. Where it is zero, there is a vertical asymptote.*

| Function      | Denominator = 0 when | Vertical asymptote(s) | Domain |
|---------------|----------------------|-----------------------|--------|
| $1/(x - 3)$   | _____                | _____                 | _____  |
| $1/(x^2 - 4)$ | _____                | _____                 | _____  |
| $1/(x^2 + 1)$ | _____                | _____                 | _____  |

The last row has no vertical asymptote at all. Explain why:

\_\_\_\_\_

$x^2 + 1$  is \_\_\_\_\_, so it is never zero for a real  $x$ .

## Part 5 — Going Further

1. True or false, with a reason: a graph may cross a horizontal asymptote. \_\_\_\_\_

2. Does  $1/x$  itself ever cross  $y = 0$ ? \_\_\_\_\_ Is that a contradiction with item 1?

\_\_\_\_\_

3. Solve  $1/x = x$ . Then say what the solutions mean on the graph.

\_\_\_\_\_

$x =$  \_\_\_\_\_ — \_\_\_\_\_

## Exit Ticket

Give the domain and both asymptotes of  $f(x) = 1/(x + 5)$ , and state its end behavior as  $x \rightarrow \infty$ .

Domain \_\_\_\_\_ VA \_\_\_\_\_ HA \_\_\_\_\_

End behavior: as  $x \rightarrow \infty$ ,  $f(x) \rightarrow$  \_\_\_\_\_

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## Parent Functions IV: $b^x$ and log base b

Algebra II Honors | Unit 1 | Day 7 | TEK A2.2.A

**Objective:** I can graph exponential and logarithmic parent functions, analyze their attributes, convert between exponential and logarithmic form, and explain why their domains and ranges are exchanged.

### Vocabulary

**Exponential function** — the variable sits in the \_\_\_\_\_. **Base** —  $b = 2, 10$ , or \_\_\_\_\_.

**Logarithm** — log base b of a answers: what \_\_\_\_\_ on b gives a? So a logarithm IS an exponent.

**Two forms** —  $b^x = a$  and log base b of  $a = x$  say the \_\_\_\_\_ two ways.

### Part 1 — Two Tables That Trade Places

|                                |    |    |   |   |   |   |
|--------------------------------|----|----|---|---|---|---|
| <b>x</b>                       | -2 | -1 | 0 | 1 | 2 | 3 |
| <b>f(x) = <math>2^x</math></b> |    |    |   |   |   |   |

|                                      |     |     |   |   |   |   |
|--------------------------------------|-----|-----|---|---|---|---|
| <b>x</b>                             | 1/4 | 1/2 | 1 | 2 | 4 | 8 |
| <b>g(x) = <math>\log_2(x)</math></b> |     |     |   |   |   |   |

State precisely what happened between the two tables: \_\_\_\_\_

Can  $2^x$  ever be 0 or negative? \_\_\_\_\_ So what is the domain of  $\log_2$ ? \_\_\_\_\_

### Part 2 — Converting Between Forms

Rewrite each in the other form. A logarithm is just an exponent asked for backwards.

| Exponential form | Logarithmic form     | Exponential form | Logarithmic form   |
|------------------|----------------------|------------------|--------------------|
| $2^x = 8$        | _____                | $10^x = 1000$    | _____              |
| _____            | log base 5 of 25 = 2 | _____            | $\log_2(1/8) = -3$ |
| $3^x = 81$       | _____                | $e^x = 1$        | _____              |

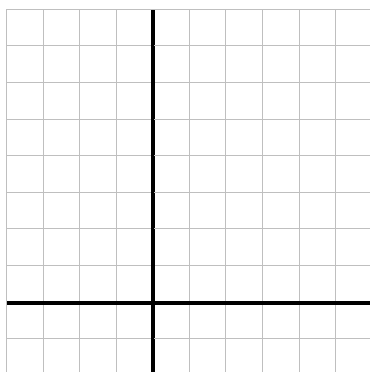
Solve by inspection:  $2^x = 32 \rightarrow x =$  \_\_\_\_\_  $\log_2(64) =$  \_\_\_\_\_  $\log(100) =$  \_\_\_\_\_

Why is log base b of 1 always 0, for any base? \_\_\_\_\_

### Part 3 — Graph Them Both

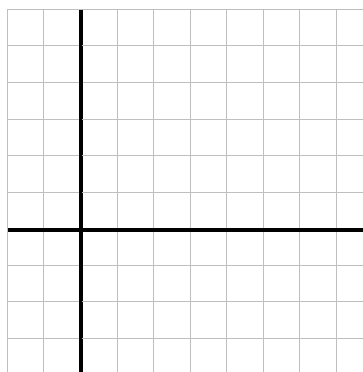
Each square is 1 unit. Draw the asymptote dashed first.

A.  $f(x) = 2^x$



HA \_\_\_\_\_ intercept \_\_\_\_\_

B.  $f(x) = \log_2(x)$



VA \_\_\_\_\_ intercept \_\_\_\_\_

#### Part 4 — Attributes Side by Side

| Attribute | $b^x$ | log base b |
|-----------|-------|------------|
| Domain    | _____ | _____      |
| Range     | _____ | _____      |
| Intercept | _____ | _____      |
| Asymptote | _____ | _____      |

Every row is swapped. Explain what relationship between the two functions causes that:

\_\_\_\_\_

They are \_\_\_\_\_ — formalized on Day 10.

#### Part 5 — Going Further

1. Which grows faster past  $x = 1$ ,  $2^x$  or  $10^x$ ? Justify with two values.

\_\_\_\_\_

2. Every exponential graph  $b^x$  passes through one shared point. Name it and explain.

\_\_\_\_\_

3. Why is log base b of a negative number undefined?

\_\_\_\_\_

No exponent on a positive base \_\_\_\_\_

#### Exit Ticket

A function has domain  $(0, \infty)$ , range all reals, x-intercept  $(1, 0)$ , and a vertical asymptote at  $x = 0$ .

Family: \_\_\_\_\_ Inverse family: \_\_\_\_\_ Rewrite  $\log_2(16) = 4$  in exponential form: \_\_\_\_\_

Name \_\_\_\_\_ Date \_\_\_\_\_ Period \_\_\_\_\_

## Comparing All Seven Parent Functions

Algebra II Honors | Unit 1 | Day 8 | TEK A2.2.A

**Objective:** I can compare all seven parent functions across their key attributes, classify each as even, odd, or neither using an algebraic test, and state end behavior in formal notation.

### Vocabulary

**Even:** \_\_\_\_\_ **Odd:** \_\_\_\_\_ Most functions are \_\_\_\_\_.

**Relative max / min** — the largest or smallest output \_\_\_\_\_, which may exist even when no overall one does.

### Part 1 — The Master Table

Complete every cell from your Day 4-7 work. This is your reference for the year.

| Function      | Domain | Range | Intercept | Asymptotes |
|---------------|--------|-------|-----------|------------|
| $\sqrt{x}$    | _____  | _____ | _____     | _____      |
| $ x $         | _____  | _____ | _____     | _____      |
| $x^3$         | _____  | _____ | _____     | _____      |
| $\sqrt[3]{x}$ | _____  | _____ | _____     | _____      |
| $1/x$         | _____  | _____ | _____     | _____      |
| $b^x$         | _____  | _____ | _____     | _____      |
| log base b    | _____  | _____ | _____     | _____      |

### Part 2 — Even, Odd, or Neither

Substitute  $-x$  into each rule. Do not decide from the picture alone.

| Function      | $f(-x)$ simplifies to | Even, odd, or neither |
|---------------|-----------------------|-----------------------|
| $ x $         | _____                 | _____                 |
| $x^3$         | _____                 | _____                 |
| $\sqrt[3]{x}$ | _____                 | _____                 |
| $1/x$         | _____                 | _____                 |
| $\sqrt{x}$    | _____                 | _____                 |
| $2^x$         | _____                 | _____                 |

$\sqrt{x}$  is "neither" for a different reason than  $2^x$ . Explain the difference:

\_\_\_\_\_  
 \_\_\_\_\_

$\sqrt[3]{-x}$  is \_\_\_\_\_ for positive  $x$ , while  $2^x$  is defined but \_\_\_\_\_.

### Part 3 — Name That Function

1. Odd, no intercepts, two asymptotes. \_\_\_\_\_
2. Even, minimum 0, domain all reals. \_\_\_\_\_
3. Domain  $(0, \infty)$ , range all reals, through  $(1, 0)$ . \_\_\_\_\_

4. Odd, domain and range all reals, and steepens as  $x$  grows. \_\_\_\_\_

5. Items 4 and its partner  $\sqrt[3]{x}$  are both odd with the same domain and range. What single piece of information separates them?

\_\_\_\_\_

One point, such as \_\_\_\_\_

#### Part 4 — Extremes on an Interval

a.  $f(x) = x^3$  on  $[-2, 2]$ : min \_\_\_\_\_ max \_\_\_\_\_      b.  $f(x) = |x|$  on  $[-3, 1]$ : min \_\_\_\_\_  
max \_\_\_\_\_

c.  $f(x) = 1/x$  on  $[1, 4]$ : min \_\_\_\_\_ max \_\_\_\_\_      d.  $f(x) = 1/x$  on  $[-1, 1]$ :  
\_\_\_\_\_

Item (d) is the interesting one. Why does a closed interval fail to guarantee a max and min here?

\_\_\_\_\_

\_\_\_\_\_

The function is \_\_\_\_\_ — there is a break at  $x = 0$ .

#### Part 5 — Going Further

1. Sort all seven by whether the domain is restricted. restricted: \_\_\_\_\_

2. Only two of the seven have a minimum. Name them and explain what they share.

\_\_\_\_\_

$\sqrt{x}$  and  $|x|$  — \_\_\_\_\_

3. Must an even function pass through the origin? Give an example or counterexample.

\_\_\_\_\_

No — \_\_\_\_\_

#### Exit Ticket

Classify  $f(x) = 1/x$  as even, odd, or neither, showing the algebra, and name one other parent function in the same class.

\_\_\_\_\_

\_\_\_\_\_

$f(-x) =$  \_\_\_\_\_ Another odd one: \_\_\_\_\_



Name \_\_\_\_\_ Date \_\_\_\_\_ Period \_\_\_\_\_

## Modeling with a Parent Function: the Open Box

Algebra II Honors | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

**Objective:** I can model a situation with a cubic function, justify its contextual domain, locate a maximum numerically, and generalize the model to any rectangular sheet.

### The Problem

A flat sheet of cardboard is **20 in. by 10 in.** A square of side  $x$  is cut from each corner and the flaps are folded up to form an open-top box.

### Part 1 — Build and Expand

Each cut removes  $x$  from BOTH ends of a side. Write each dimension, then the volume.

height \_\_\_\_\_ length \_\_\_\_\_ width \_\_\_\_\_

$V(x) =$  \_\_\_\_\_

Now expand to standard form. Show the multiplication.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

$V(x) =$  \_\_\_\_\_ Degree \_\_\_\_\_ Family \_\_\_\_\_

What are the three zeros of the expanded form, and which are physically meaningful?

Zeros at  $x =$  \_\_\_\_\_ meaningful: \_\_\_\_\_

### Part 2 — Justify the Domain

Why  $x > 0$ ? \_\_\_\_\_

Why  $x < 5$  and not  $x < 10$ ? \_\_\_\_\_

Contextual domain \_\_\_\_\_ Open or closed, and why? \_\_\_\_\_

### Part 3 — Locate the Maximum

Compute  $V(x)$ . Round to the nearest hundredth where needed.

| $x$    | 1 | 2 | 2.1 | 2.11 | 2.2 | 3 | 4 |
|--------|---|---|-----|------|-----|---|---|
| $V(x)$ |   |   |     |      |     |   |   |

Average rate of change on  $[1, 2]$ : \_\_\_\_\_ on  $[2, 3]$ : \_\_\_\_\_

The sign flips between those intervals. What does that tell you, and why is it stronger evidence than just reading the largest table value?

\_\_\_\_\_

\_\_\_\_\_

The volume stops rising and starts falling, so a maximum lies \_\_\_\_\_.

Maximum  $\approx$  \_\_\_\_\_ at  $x \approx$  \_\_\_\_\_ Box  $\approx$  \_\_\_\_\_

Range on this domain: \_\_\_\_\_ Why is 0 excluded but 192.45 included?

\_\_\_\_\_

## Part 4 — Generalize

Now let the sheet be **L by W** with W the shorter side, and squares of side  $x$  cut from the corners.

$V(x) =$  \_\_\_\_\_ Contextual domain \_\_\_\_\_

Explain why the upper bound depends on W and never on L:

---

---

The shorter side reaches zero first, so it \_\_\_\_\_.

Apply it: a 16 by 8 sheet gives domain \_\_\_\_\_ a 30 by 30 sheet gives domain \_\_\_\_\_

## Part 5 — Going Further

1. For the 16 by 8 sheet, compute  $V(1)$  and  $V(2)$ . Show the arithmetic.

---

---

$V(1) =$  \_\_\_\_\_  $V(2) =$  \_\_\_\_\_

2. For a square sheet  $s$  by  $s$ , write  $V(x)$  and its domain.  $V(x) =$  \_\_\_\_\_ domain \_\_\_\_\_

3. Does doubling both sheet dimensions double the maximum volume? Predict, then justify.

---

---

No — \_\_\_\_\_

4. A sheet is 18 by 6. Find the domain, then compute  $V$  at the midpoint of that domain.

---

---

Domain \_\_\_\_\_ midpoint  $x =$  \_\_\_\_\_  $V(1.5) =$  \_\_\_\_\_

5. Is the maximum always at the midpoint of the domain? Test against the 20 by 10 sheet.

---

---

No — \_\_\_\_\_

## Exit Ticket

A 12 in. by 12 in. sheet has squares of side  $x$  cut from each corner.

$V(x) =$  \_\_\_\_\_ Domain \_\_\_\_\_  $V(2) =$  \_\_\_\_\_

Explain the upper bound in one sentence: \_\_\_\_\_

Inverses from Graphs and Tables

Algebra II Honors | Unit 1 | Day 10 | TEK A2.2.B

**Objective:** I can find the inverse of a relation from a table or graph, use  $f^{-1}(x)$  notation correctly, justify why domain and range exchange, and predict whether an inverse will be a function.

Vocabulary

- Inverse** — the relation that \_\_\_\_\_ the original. Written \_\_\_\_\_, which does \_\_\_\_\_ mean  $1/f(x)$ .
- Building it** — in a table or a set of pairs, \_\_\_\_\_. On a graph, reflect across \_\_\_\_\_.
- One-to-one** — no output is repeated. Test with a \_\_\_\_\_ line: if it meets the graph twice, the function is not one-to-one.

Part 1 — Inverses from Pairs and Tables

- a.  $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$   
 $f^{-1} = \{ \text{_____} \}$
- b. Complete the inverse table.

|      |    |    |   |    |   |
|------|----|----|---|----|---|
| x    | -4 | 0  | 2 | 3  | 7 |
| f(x) | -6 | -1 | 4 | -7 | 7 |

|                     |    |    |   |    |   |
|---------------------|----|----|---|----|---|
| x                   | -6 | -1 | 4 | -7 | 7 |
| f <sup>-1</sup> (x) |    |    |   |    |   |

- c. Which point in  $f$  is unchanged by the swap, and what is special about it? \_\_\_\_\_

Part 2 — Why Domain and Range Exchange

- Argue this in general, not from one example.
- If  $(a, b)$  belongs to  $f$ , then \_\_\_\_\_ belongs to  $f^{-1}$  by definition of the swap.
- So every input of  $f^{-1}$  was an \_\_\_\_\_ of  $f$ . Therefore:
- domain of  $f^{-1}$  = \_\_\_\_\_ range of  $f^{-1}$  = \_\_\_\_\_

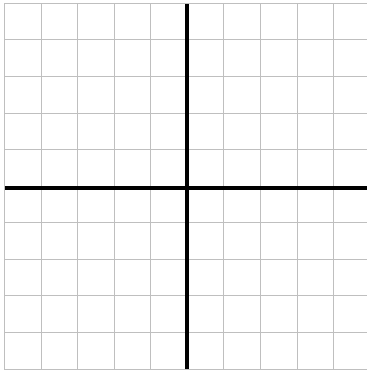
| f          | Domain of f   | Range of f    | Domain of f <sup>-1</sup> | Range of f <sup>-1</sup> |
|------------|---------------|---------------|---------------------------|--------------------------|
| $x^3$      | all reals     | all reals     | _____                     | _____                    |
| $2^x$      | all reals     | $(0, \infty)$ | _____                     | _____                    |
| $\sqrt{x}$ | $[0, \infty)$ | $[0, \infty)$ | _____                     | _____                    |

Row 2 explains something from Day 7. What? \_\_\_\_\_

Part 3 — Graph a Pair

Each square is 1 unit. Plot  $f$ , then  $f^{-1}$ , then draw  $y = x$ .

$$f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$$



**Record it.**

Reflection line: \_\_\_\_\_

Fixed points: \_\_\_\_\_

If  $(a, b)$  is on  $f$ , then on  $f^{-1}$ :

\_\_\_\_\_

Is  $f$  one-to-one here? \_\_\_\_\_

$$f^{-1} = \{ \text{_____} \}$$

### Part 4 — Will the Inverse Be a Function?

*Apply the horizontal line test to the original, before building the inverse.*

a.  $g = \{ (2, 6), (4, 6), (5, 1) \}$ . Write  $g^{-1}$ : \_\_\_\_\_

Is  $g^{-1}$  a function? \_\_\_\_\_ Why? \_\_\_\_\_

What was true of  $g$  that caused it? \_\_\_\_\_

b. Which of these parent functions have inverses that are functions? Circle:

$x^3$   $x^2$   $2^x$   $|x|$   $\sqrt[3]{x}$  Answer: \_\_\_\_\_

What do the ones that fail have in common? \_\_\_\_\_

### Part 5 — Going Further

1. Does  $f^{-1}(x)$  mean  $1/f(x)$ ? Give a specific counterexample with numbers.

\_\_\_\_\_

\_\_\_\_\_

For  $f(x) = x + 3$ :  $f^{-1}(x) = \text{_____}$  but  $1/f(x) = \text{_____}$  — different functions.

2. A function is its own inverse. What must be true of its graph? \_\_\_\_\_

3. Can a function have two different inverses? Argue yes or no.

\_\_\_\_\_

No — \_\_\_\_\_

### Exit Ticket

For  $h = \{ (1, 4), (2, 9), (5, 0) \}$ , write  $h^{-1}$ , give its domain and range, and say whether it is a function with a reason.

$h^{-1} = \{ \text{_____} \}$  Domain \_\_\_\_\_ Range \_\_\_\_\_

A function? \_\_\_\_\_

Name \_\_\_\_\_ Date \_\_\_\_\_ Period \_\_\_\_\_

## Writing the Inverse of a Linear Function

Algebra II Honors | Unit 1 | Day 11 | TEK A2.2.B

**Objective:** I can write the inverse of a linear function algebraically, derive the inverse of the general form  $f(x) = mx + b$ , and prove that a line and its inverse have reciprocal slopes.

### Warm-Up — This Is a Skill You Have

*Finding an inverse is solving a literal equation for the other variable. Nothing new.*

Solve  $y = 3x + 6$  for  $x$ :  $x =$  \_\_\_\_\_ Solve  $y = mx + b$  for  $x$ :  $x =$  \_\_\_\_\_

### The Procedure

1. Replace  $f(x)$  with \_\_\_\_\_.
2. \_\_\_\_\_.
3. Solve for  $y$  and rename it \_\_\_\_\_.

### Part 1 — Find Each Inverse

*Show the swap-and-solve steps, not just the answer.*

a.  $f(x) = 2x + 3$

\_\_\_\_\_

$f^{-1}(x) =$  \_\_\_\_\_

b.  $f(x) = -2x + 8$

\_\_\_\_\_

$f^{-1}(x) =$  \_\_\_\_\_

c.  $f(x) = (1/2)x - 6$

\_\_\_\_\_

$f^{-1}(x) =$  \_\_\_\_\_

d.  $f(x) = (3x - 1)/4$

\_\_\_\_\_

\_\_\_\_\_

$f^{-1}(x) =$  \_\_\_\_\_

### Part 2 — Derive the General Case

Now do it once for every linear function at the same time. Let  $f(x) = mx + b$ , with  $m \neq 0$ .

\_\_\_\_\_

\_\_\_\_\_

$f^{-1}(x) =$  \_\_\_\_\_ which in slope-intercept form is  $y =$  \_\_\_\_\_

Slope of  $f$  is \_\_\_\_\_ and slope of  $f^{-1}$  is \_\_\_\_\_, so the slopes are \_\_\_\_\_. That is now proved, not just noticed.

Why did the derivation require  $m \neq 0$ ? What kind of line is  $m = 0$ , and what goes wrong?

\_\_\_\_\_

A \_\_\_\_\_ line. Every input maps to one output, so the inverse \_\_\_\_\_.

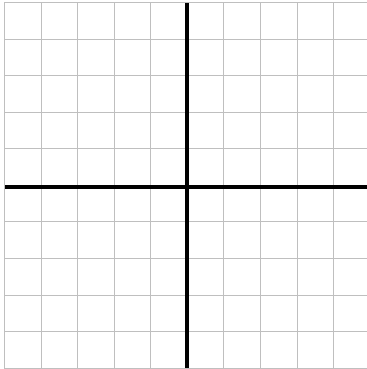
### Part 3 — Check Two Ways

Use  $f(x) = 2x + 3$  and  $f^{-1}(x) = (x - 3)/2$ .

|             |    |   |   |   |
|-------------|----|---|---|---|
| <b>x</b>    | -1 | 0 | 1 | 2 |
| <b>f(x)</b> | 1  | 3 | 5 | 7 |

|                          |   |   |   |   |
|--------------------------|---|---|---|---|
| <b>x</b>                 | 1 | 3 | 5 | 7 |
| <b>f<sup>-1</sup>(x)</b> |   |   |   |   |

Graph  $f$ ,  $f^{-1}$ , and  $y = x$ .



Record it.

slope of  $f$  \_\_\_\_\_ slope of  $f^{-1}$  \_\_\_\_\_

y-int of  $f$  \_\_\_\_\_ of  $f^{-1}$  \_\_\_\_\_

Where do  $f$  and  $f^{-1}$  intersect?

\_\_\_\_\_

### Part 4 — Self-Inverse Lines

A function is self-inverse when  $f^{-1}(x) = f(x)$ . Using  $f(x) = mx + b$ , find every linear function with this property.

\_\_\_\_\_

Need  $(x - b)/m = mx + b$ , which forces  $m =$  \_\_\_\_\_

If  $m = 1$ ,  $b$  must be \_\_\_\_\_, giving \_\_\_\_\_. If  $m = -1$ ,  $b$  may be \_\_\_\_\_, giving  $f(x) = -x + b$ .

Check one: is  $f(x) = -x + 6$  its own inverse? Test with  $x = 2$ .

$f(2) =$  \_\_\_\_\_  $f(4) =$  \_\_\_\_\_ so yes, it returns the start. \_\_\_\_\_

### Part 5 — In Context

Celsius to Fahrenheit:  $F(c) = (9/5)c + 32$ .

$F^{-1}(x) =$  \_\_\_\_\_ In words, the inverse \_\_\_\_\_

Is there a temperature where the two scales agree? Set  $(9/5)c + 32 = c$  and solve.

\_\_\_\_\_

$c =$  \_\_\_\_\_ — \_\_\_\_\_

### Exit Ticket

Write the inverse of  $f(x) = 6x - 12$ , verify with one input, and state the relationship between the two slopes.

$f^{-1}(x) =$  \_\_\_\_\_  $f(3) =$  \_\_\_\_\_ and  $f^{-1}(6) =$  \_\_\_\_\_

Slopes: \_\_\_\_\_

## When Is an Inverse a Function?

Algebra II Honors | Unit 1 | Day 12 | TEK A2.2.C

**Objective:** I can determine whether the inverse of a function is a function, choose a domain restriction that makes it one, and apply this to a quadratic whose vertex is not at the origin.

### Vocabulary

**One-to-one** — every output comes from exactly one input. Detected by the \_\_\_\_\_.

**Key fact** — the inverse of a function is always a \_\_\_\_\_, but a \_\_\_\_\_ only when the original is one-to-one.

**Domain restriction** — deliberately shrinking the domain to force one-to-one behavior. Restricting the domain of  $f$  restricts the \_\_\_\_\_ of  $f^{-1}$ .

### Part 1 — Why $f(x) = x^2$ Fails

| x            | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
|--------------|----|----|----|---|---|---|---|
| $f(x) = x^2$ |    |    |    |   |   |   |   |

Repeated outputs: \_\_\_\_\_ So  $f$  is \_\_\_\_\_

Swapping produces both  $(1, 1)$  and  $(1, -1)$ . State precisely which test the inverse fails and why:

\_\_\_\_\_

\_\_\_\_\_

The \_\_\_\_\_ — the input 1 would need two outputs.

Solving  $y = x^2$  for  $x$  gives  $x = \pm \sqrt{y}$ , and the \_\_\_\_\_ is the whole problem.

### Part 2 — Restrict and Repair

| Restriction on $x^2$ | Half of the parabola kept | Resulting $f^{-1}(x)$ | Range of $f^{-1}$ |
|----------------------|---------------------------|-----------------------|-------------------|
| $x \geq 0$           | _____                     | _____                 | _____             |
| $x \leq 0$           | _____                     | _____                 | _____             |

Both restrictions work. Is one more correct than the other? \_\_\_\_\_

### Part 3 — A Vertex That Is Not at the Origin

Now take  $g(x) = (x - 2)^2$ . The same idea applies, but you must locate the turning point first.

Vertex of  $g$ : \_\_\_\_\_ So the natural restriction is \_\_\_\_\_

Now find the inverse. Swap and solve for  $y$ , keeping track of the  $\pm$ .

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

$$x = (y - 2)^2 \rightarrow \text{_____} \rightarrow y = \text{_____}$$

With the restriction  $x \geq 2$  on  $g$ , the inverse is  $g^{-1}(x) = \text{_____}$  Domain of  $g^{-1}$  \_\_\_\_\_

Why does the 2 appear OUTSIDE the radical in the inverse when it was inside the square in g?

---

---

Undoing happens in \_\_\_\_\_: take the root first, then add 2.

#### Part 4 — Which Need No Restriction?

| Function | One-to-one? | Restriction needed? | Inverse |
|----------|-------------|---------------------|---------|
| $x^2$    | _____       | _____               | _____   |
| $x^3$    | _____       | _____               | _____   |
| $2^x$    | _____       | _____               | _____   |
| $ x $    | _____       | _____               | _____   |
| $1/x$    | _____       | _____               | _____   |

State the pattern as a rule about increasing and decreasing:

---

A function that only increases or only decreases is \_\_\_\_\_; one that turns around is not.

#### Part 5 — Going Further

1. Restrict  $h(x) = (x + 3)^2$  so its inverse is a function, then write the inverse.

---

---

Restrict \_\_\_\_\_  $h^{-1}(x) =$  \_\_\_\_\_

2. Explain why  $1/x$  is its own inverse, using the swap.

---

3. Is there a restriction of  $|x|$  whose inverse is a function? Give one. \_\_\_\_\_

#### Exit Ticket

Consider  $f(x) = (x - 5)^2$  with no restriction.

Is its inverse a function? \_\_\_\_\_ Give a restriction that fixes it: \_\_\_\_\_

With that restriction,  $f^{-1}(x) =$  \_\_\_\_\_ Explain in one sentence why the restriction is needed:

---



Name \_\_\_\_\_ Date \_\_\_\_\_ Period \_\_\_\_\_

## Composition to Verify Inverses

Algebra II Honors | Unit 1 | Day 13 | TEK A2.2.D

**Objective:** I can compose two functions, use  $f(g(x)) = x$  and  $g(f(x)) = x$  to prove they are inverses, and state the domain restriction a non-linear pair requires.

### Vocabulary

**Composition** — one function inside another,  $f(g(x))$  or  $(f \circ g)(x)$ . Evaluate from the \_\_\_\_\_.

**The inverse test** —  $f$  and  $g$  are inverses only when \_\_\_\_\_ AND \_\_\_\_\_. One direction is not enough.

**Order matters** — in general  $f(g(x))$  \_\_\_\_\_ the same as  $g(f(x))$ .

### Part 1 — Warm Up and Prove

Let  $f(x) = 3x - 5$  and  $g(x) = (x + 5)/3$ .

$f(4) =$  \_\_\_\_\_  $g(7) =$  \_\_\_\_\_  $f(g(4)) =$  \_\_\_\_\_  $g(f(4)) =$  \_\_\_\_\_

Now prove it in general, both directions.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

$f(g(x)) = 3 \cdot (x+5)/3 - 5 =$  \_\_\_\_\_  $g(f(x)) = ((3x-5)+5)/3 =$  \_\_\_\_\_ Inverses? \_\_\_\_\_

Show that order matters in general: for  $f(x) = x + 1$  and  $h(x) = 2x$ , find both compositions.

$f(h(x)) =$  \_\_\_\_\_  $h(f(x)) =$  \_\_\_\_\_ Equal? \_\_\_\_\_

### Part 2 — A Pair That Looks Right and Is Not

Test  $f(x) = 2x + 1$  and  $g(x) = 2x - 1$ . Compute, do not assume.

\_\_\_\_\_  
\_\_\_\_\_

$f(g(x)) =$  \_\_\_\_\_ Equal to  $x$ ? \_\_\_\_\_ Actual inverse of  $f$ : \_\_\_\_\_

State the lesson in your own words:

\_\_\_\_\_

Changing a sign is not undoing a function — you must reverse \_\_\_\_\_.

### Part 3 — Non-Linear Pairs and Restrictions

Test  $f(x) = x^2$  and  $g(x) = \sqrt{x}$ . Try  $x = 9$  and then  $x = -9$ .

$f(g(9)) =$  \_\_\_\_\_  $g(f(9)) =$  \_\_\_\_\_  $g(f(-9)) =$  \_\_\_\_\_

So  $g(f(x)) = x$  holds only when \_\_\_\_\_. In general  $g(f(x)) = \sqrt{x^2} =$  \_\_\_\_\_, not  $x$ .

Write the complete, correct statement of the inverse relationship for this pair:

\_\_\_\_\_  
\_\_\_\_\_

f and g are inverses \_\_\_\_\_

Now test  $f(x) = x^3$  and  $g(x) = \sqrt[3]{x}$  the same way, at  $x = -8$ .

$g(f(-8)) =$  \_\_\_\_\_ Restriction needed? \_\_\_\_\_ Why the difference from the squaring pair?  
\_\_\_\_\_

#### Part 4 — Error Analysis

A student is given  $f(x) = 2(x - 3)^2 + 4$  and claims  $f^{-1}(x) = \sqrt{x - 3} + 4$ .

Test the claim at  $x = 6$ : the claimed inverse gives \_\_\_\_\_, and f of that is \_\_\_\_\_.

Undo the steps in reverse: subtract 4, divide by 2, take the root, add 3.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Correct:  $f^{-1}(x) =$  \_\_\_\_\_ Restriction making it a function: \_\_\_\_\_

Where does the  $\pm$  come from, in terms of Day 12? \_\_\_\_\_

#### Part 5 — Going Further

1. Verify  $f(x) = 5x$  and  $g(x) = x/5$  are inverses, both directions. \_\_\_\_\_

2. Show  $f(x) = 1/x$  is its own inverse by composing it with itself.

\_\_\_\_\_  
\_\_\_\_\_

$f(f(x)) = 1/(1/x) =$  \_\_\_\_\_ Other self-inverse functions: \_\_\_\_\_

3. Prove in general that  $f(x) = mx + b$  and  $f^{-1}(x) = (x - b)/m$  compose to x.

\_\_\_\_\_  
\_\_\_\_\_

4. If  $f(g(x)) = x$  but  $g(f(x)) \neq x$ , can they still be called inverses? Explain.

\_\_\_\_\_

No — \_\_\_\_\_

#### Exit Ticket

Decide whether  $f(x) = 4x - 8$  and  $g(x) = (x + 8)/4$  are inverses. Show both compositions, then explain why one direction would not be sufficient proof.

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$f(g(x)) =$  \_\_\_\_\_  $g(f(x)) =$  \_\_\_\_\_ Inverses? \_\_\_\_\_